CONCEPTUAL SERIES
Load Balancing

Name: ______________________ Cohort/Batch: ______________________ Date: ______________________ Score: ________

LOAD BALANCING PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Distributed Systems

TOTAL: 10 QUESTIONS

Q1 What is Load Balancing and what problem in Distributed Systems does it solve? Threat Model

Q2 What are the core components or concepts in Load Balancing? Architecture

Q3 How does Load Balancing compare to its main alternatives? Configuration

Q4 How do you get started with Load Balancing for a new project? Hardening

Q5 What configuration options most affect Load Balancing's behavior in production? SSO / IdP

Q6 How does Load Balancing handle reliability and high availability? Observability

Q7 What observability does Load Balancing provide out of the box? Lifecycle

Q8 What are common performance bottlenecks in Load Balancing? Compliance

Q9 What security best practices apply when running Load Balancing? Testing

Q10 What mistakes do teams commonly make when adopting Load Balancing? Tracing

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Load Balancing is a tool or platform

Mitigation

Load Balancing is a tool or platform that automates or simplifies a specific concern in the Distributed Systems domain, reducing manual effort and operational complexity for engineering teams.

A2 Load Balancing has key building blocks that

Primitives

Load Balancing has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A3 Load Balancing trades off simplicity against feature

Hardening

Load Balancing trades off simplicity against feature richness differently than its alternatives. Choose Load Balancing when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A4 Install Load Balancing via its package manager

Gotchas

Install Load Balancing via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A5 Production-critical settings typically include concurrency

Federation

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A6 Load Balancing provides HA through leader election,

Observability

Load Balancing provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A7 Load Balancing exposes metrics (Prometheus endpoint or

Automation

Load Balancing exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A8 Performance bottlenecks typically stem from misconfigured

Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A9 Run Load Balancing with the principle of

Assessment

Run Load Balancing with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A10 Common pitfalls include skipping the documentation and

Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.